# Lesson Plan • Geometry 6.4 Triangle Midsegment Theorem

Wed., Nov. 4 • 4th hour • 10:40 • Observer: Mr. Noble • Feedback on #9, #14, #15

| Min | What | Notes |
|---|---|---|
| **0–5** | **Bell ringer (ACT)** | Midpoint of (−4, 7) and (10, −3). Answer: A (3, 2). B = subtracted, C = didn't divide by 2, D = sign error. |
| **5–15** | **Discovery in pairs** | One triangle sheet per pair. Circulate; make sure both partners write. |
| **15–25** | **Debrief + model** | Fill the table slide, reveal the key, students state the rule, then theorem + Examples 1–3. |
| **25–40** | **Practice on Chromebooks** | Google Classroom → 6.4. Watch the Practice Results sheet; help low scores first. |
| **40–45** | **Exit ticket + closure** | Hand out exit tickets. Collect. Students finish: “A midsegment is ___ and ___.” |

## Before class

- [ ] Print the discovery sheets (one triangle per pair) and the exit tickets (cut in half).
- [ ] Open the slides on the TV computer and click Present.
- [ ] Post the 6.4 practice link in 4th hour's Google Classroom (bonus is already included).
- [ ] Enter the bell ringer in edia.app.
- [ ] Have the Practice Results sheet open on your computer.

## Worked examples (answers)

Ex. 1: BC = 26 → DE = 13. DE = 7.5 → BC = 15.

Ex. 2: DE = 3x − 2, BC = 4x + 6 → 2(3x − 2) = 4x + 6 → x = 5, DE = 13.

Ex. 3: A(−2, 4), B(−6, −4), C(4, −2): D(−4, 0), E(1, 1); both slopes 1/5; DE = √26, BC = 2√26.

# Discovery Answer Key (teacher)

Every triangle: DE is parallel to BC (same slope) and half as long.

| # | A | B | C | D | E | slope DE | slope BC | DE | BC |
|---|---|---|---|---|---|---|---|---|---|
| 1 | (2, 10) | (0, 0) | (6, 8) | (1, 5) | (4, 9) | 4/3 | 4/3 | 5 | 10 |
| 2 | (−4, 6) | (−6, −2) | (2, −8) | (−5, 2) | (−1, −1) | −3/4 | −3/4 | 5 | 10 |
| 3 | (4, −2) | (−4, 4) | (4, 10) | (0, 1) | (4, 4) | 3/4 | 3/4 | 5 | 10 |
| 4 | (−8, 0) | (0, 8) | (6, 0) | (−4, 4) | (−1, 0) | −4/3 | −4/3 | 5 | 10 |
| 5 | (0, −6) | (−8, 2) | (2, 2) | (−4, −2) | (1, −2) | 0 | 0 | 5 | 10 |
| 6 | (10, 6) | (2, −4) | (2, 8) | (6, 1) | (6, 7) | undefined | undefined | 6 | 12 |

## Early-finisher extension (F = midpoint of BC; DF vs. AC)

#1: F = (3, 4) DF = √5 AC = 2√5

#2: F = (−2, −5) DF = √58 AC = 2√58

#3: F = (0, 7) DF = 6 AC = 12

#4: F = (3, 4) DF = 7 AC = 14

#5: F = (−3, 2) DF = √17 AC = 2√17

#6: F = (2, 2) DF = √17 AC = 2√17

## Debrief questions

- What did every group find about the slopes? What does equal slope tell us?
- What did every group find about the lengths?
- Triangle 5 has a horizontal side and triangle 6 a vertical one. Did the pattern still hold?
- Put it in one sentence: a midsegment is _____ to the third side and _____ as long.

# Exit Ticket Answer Key (teacher)

1. DE = 9 (half of 18)
2. $2(2x + 1) = 5x - 4 \rightarrow 4x + 2 = 5x - 4 \rightarrow x = 6$. DE = 13 (BC = 26).
3. DE connects the midpoints of two sides of the triangle (AB and AC).

## What wrong answers tell you

• #1 answered 36: doubled instead of halved. Mixed up which segment is shorter.
• #2 answered x = 5/3 (from 2x + 1 = 5x − 4): set them equal, forgot "half as long."
• #2 answered x = 6 but DE = 26: found x, then gave BC instead of DE.
• #3 "it's in the middle" or "it's parallel": describes a property, not the definition.

## Sort for tomorrow

• 3/3: ready to move on.
• #1 right, #2 wrong: reteach setting up the equation with the factor of 2.
• #1 wrong: reteach the theorem itself with a picture before any algebra.